AI-Assisted Kidney Stone Detection & Classification Report

Research Examination ID: TEST_EXAM_999 | Date: 2026-11-10 21:08:41

1. Image & Study Metadata

Parameter	Value
Examination ID	TEST_EXAM_999
Modality	CT Scan
Image Format	standard
DICOM Calibrated	No (Standard 2D Image)

2. Individual Deep Learning Classification Models

Model Architecture	Status	Predicted Class	Confidence	Stone Prob.	Non-Stone Prob.	Latency
MOBILENET V2	Trained & Loaded	**Stone**	94.0%	94.0%	6.0%	12.5 ms

3. Multi-Model Ensemble Analysis

Ensemble Decision	**Stone** (94.0% confidence)
Stone Probability	94.0%
Non-Stone Probability	6.0%
Contributing Classifiers	mobilenet_v2
Normalized Model Weights	{'mobilenet_v2': 1.0}
Model Disagreement	Full Consensus

4. YOLOv8 Object Detection (Localization)

Detection model not trained or unavailable. Stone localization bounding-box annotations required.

5. Physical Measurements & Calibrations

Status: None

MANDATORY MEDICAL & RESEARCH DISCLAIMER:
This report was generated by an academic research artificial intelligence system. The classification outputs, probability scores, and ensemble results are algorithmic estimates and do NOT constitute a medical diagnosis, clinical radiologist verification, or treatment recommendation. Model confidence must not be interpreted as an absolute probability of clinical disease. Clinical diagnosis requires thorough review by a qualified radiologist or physician.